

DLA - Development of the Nervous System – Part 2

Clinical correlates

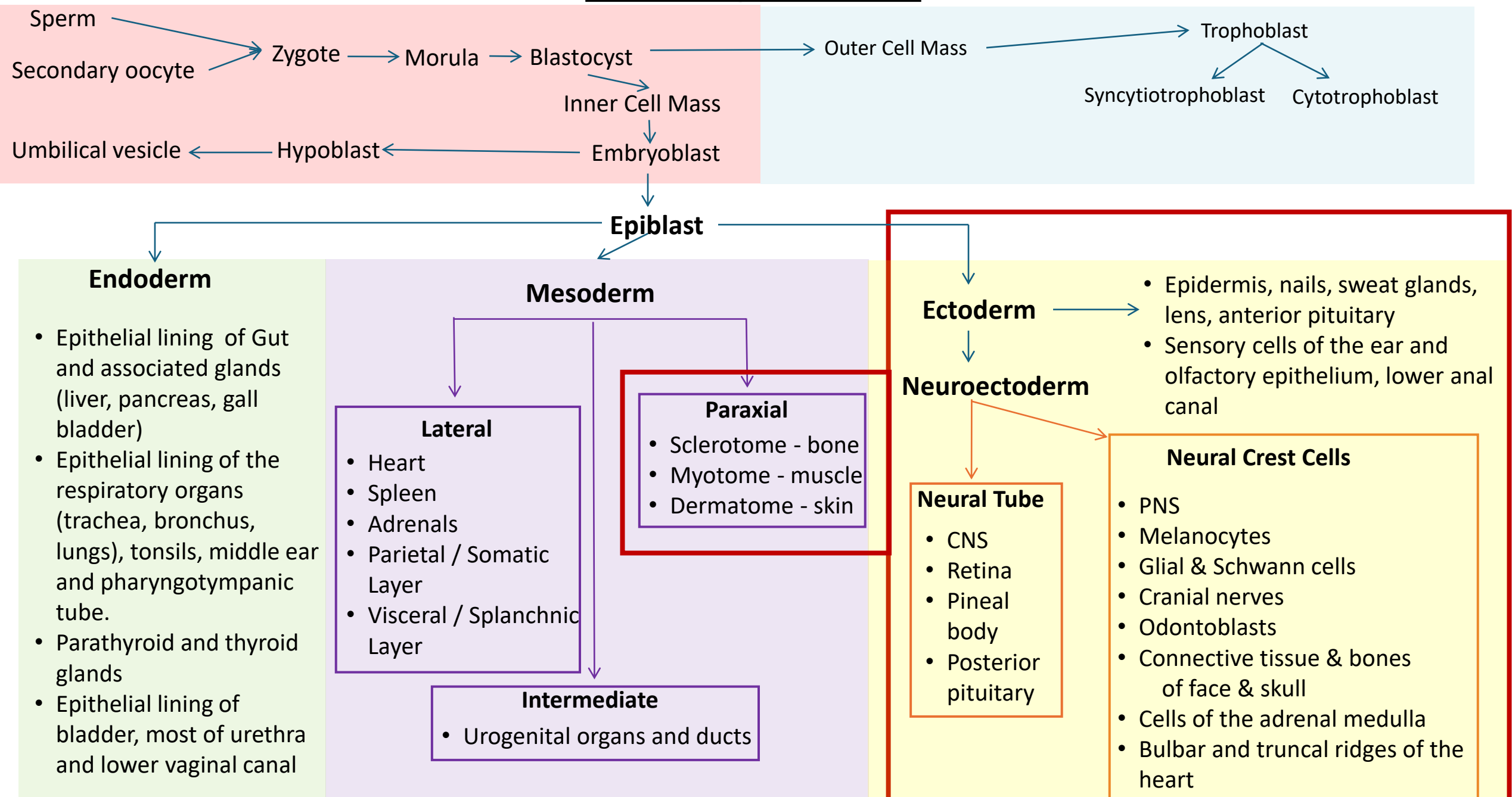
Objectives

SOM.MKII.BPM1.1.MSK.1.A NAT.1103	List common causes of neural tube defects.
SOM.MKII.BPM1.1.MSK.1.A NAT.1104	Describe the embryological processes that fail and subsequent anatomical defect and severity of; spina bifida occulta, spina bifida cystica with meningocele, spina bifida cystica with meningomyelocele and spina bifida with myeloschisis.
SOM.MKII.BPM1.1.MSK.1.A NAT.1105	Describe the embryological formation and clinical presentation of chordoma.

Special note

Some slides have a green background, this will not be covered in the video recording but are put so you have the information available

The Big picture



The journey of development

- Fertilization to gastrulation - Some primordial cell differentiation happens - **Weeks 1-3**

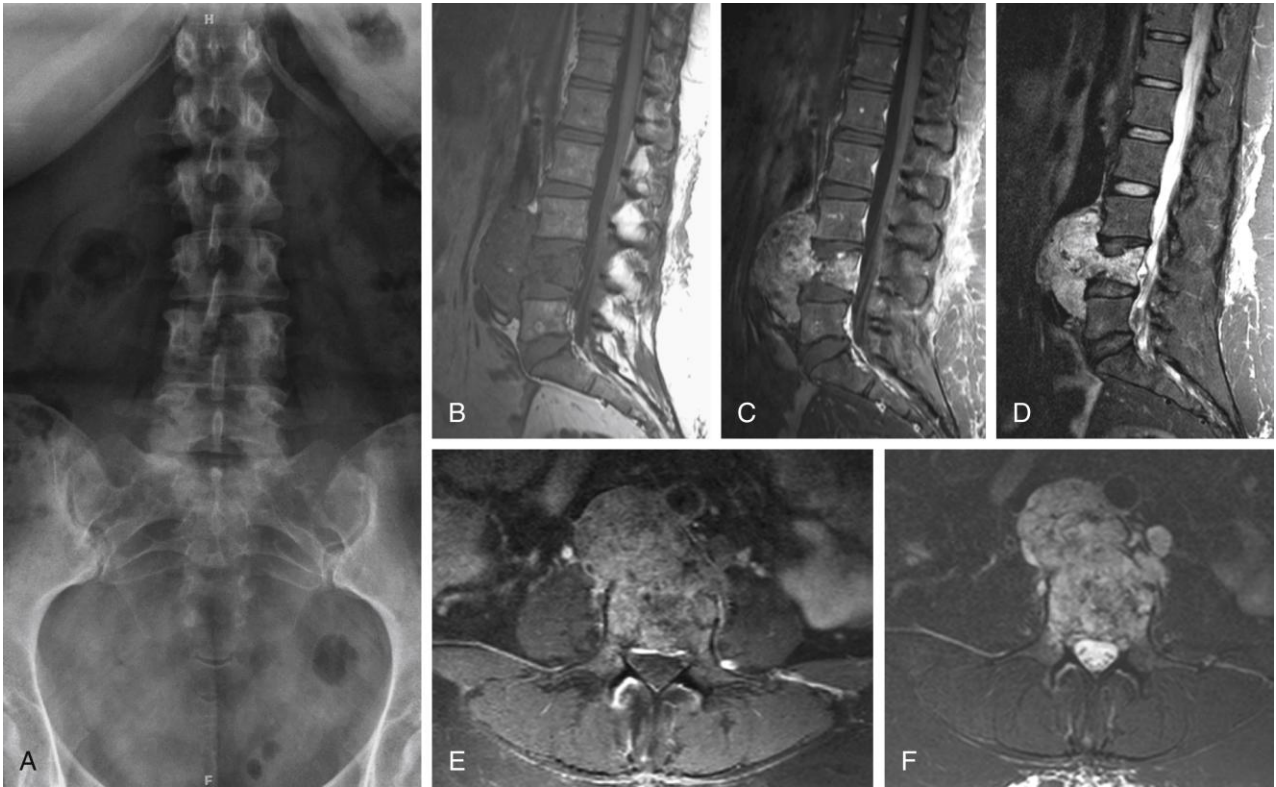
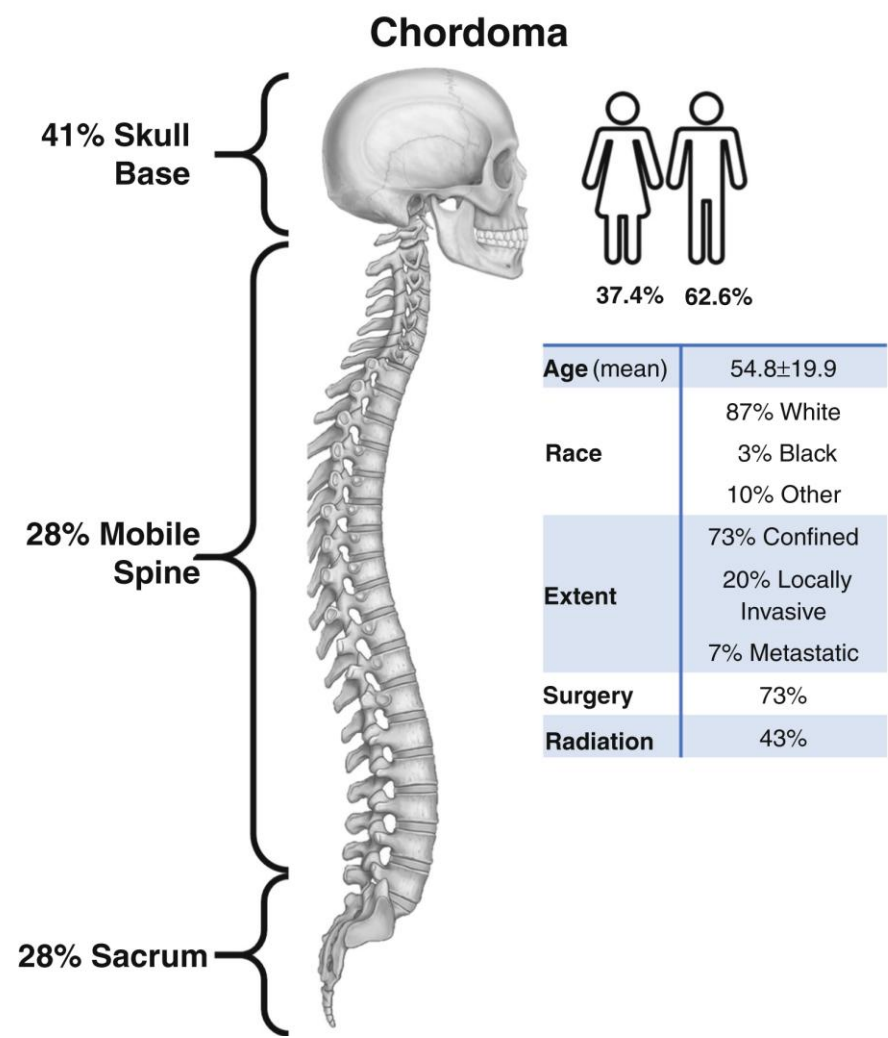
Organogenesis – Weeks 4 to 8

Everything happens synchronously with many simultaneous events

- Formation of the neural plate – day 18 – 21 (**Week 3**)
 - Tissues start to differentiate into organ primordia – day 22 - 28
 - Folding turns the embryo from a disc into a tube-inside-a-tube rendering its 3D form – day 22 – 28
 - Lateral folding turns the endoderm into the gut tube
 - Paraxial mesoderm differentiate into somites - day 22 – 30
 - Somites start differentiating into the 3 components – day 24 - 30
 - Formation of the neural tube – Neurulation – day 22 – 30
 - Neural crest cell migration – Day 22 - 30
 - Development of the spinal cord and nerves – Day 24 - 31
 - Development of the vertebrae – Day 30 - 33
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- Week 4**
- Week 5**

Chordoma

- Remnants of the notochord may form tumors

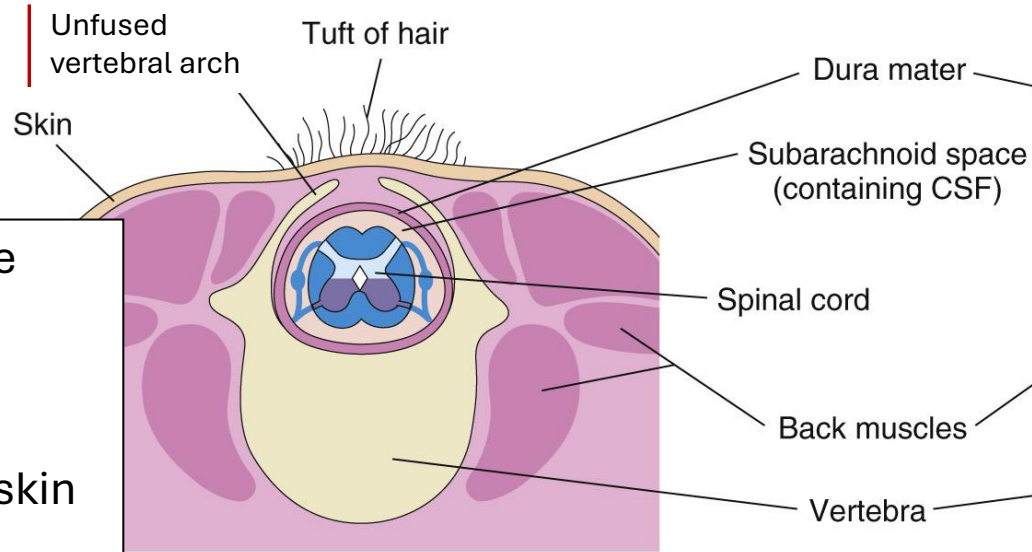


Neural tube defects (NTD's)

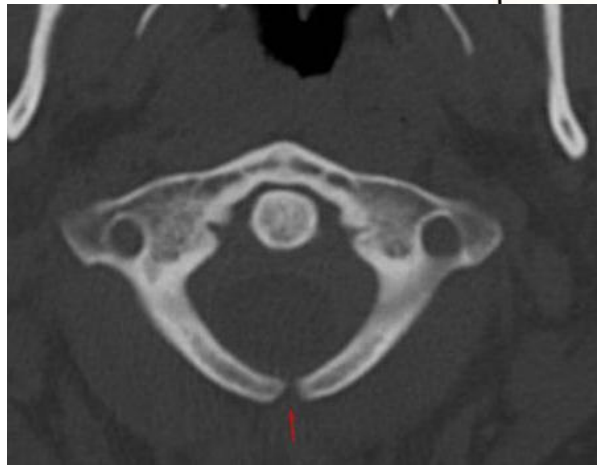
- Abnormalities of neurulation
- Occur during weeks 3-4
- May be **open** (exposed to surface) or **closed** (covered with skin)
 - Open is more severe
- May be localized or total
- Causes:
 - Genetic and environmental factors
 - Folic acid deficiency
 - Aminopterin – immunosuppressive drug used in chemotherapy
 - Carbamazepine – seizure medication
 - Isotretinoin (13- *cis* -retinoic acid) – acne treatment
- Relevant to topic
 - Spina bifida

Types of spina bifida

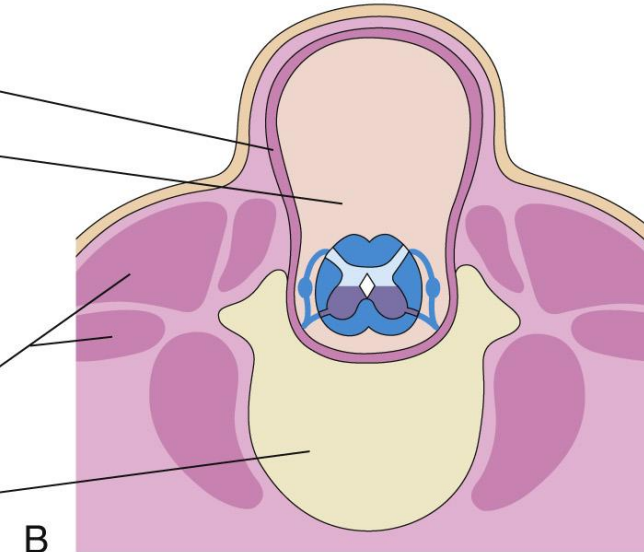
Occulta



No bulge over the bony defect
No neurological deficits
A tuft of hairs or skin dimple may be present (most common in the lumbosacral area)



Cystica with meningocele



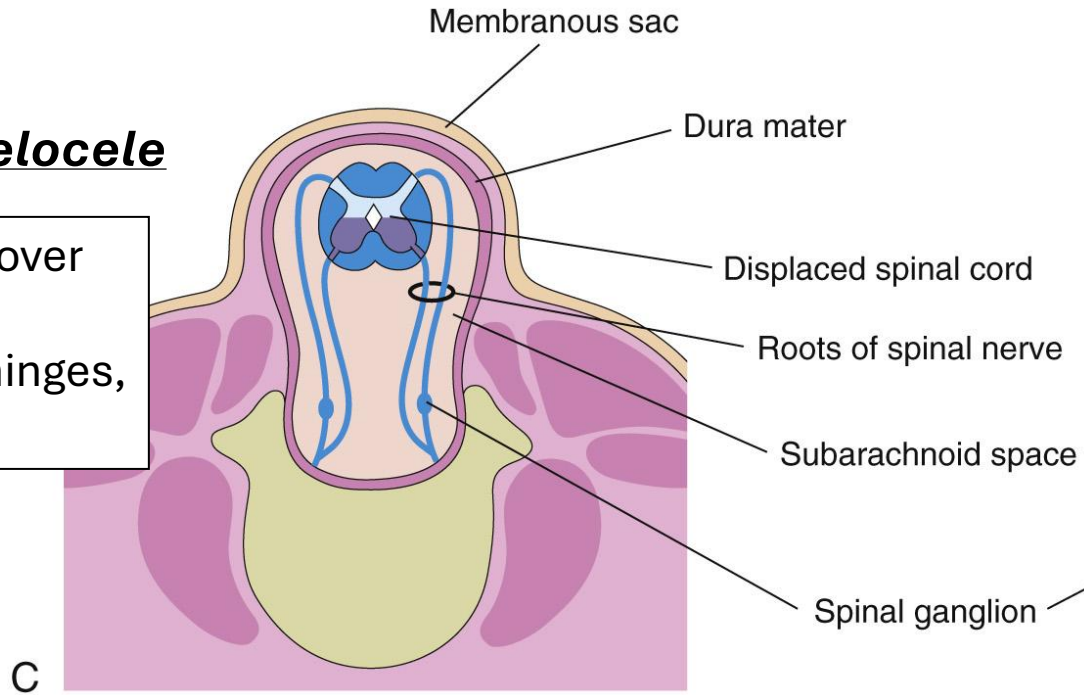
Bulge is seen over the defect
Contains meninges & CSF in the subarachnoid space



Types of spina bifida

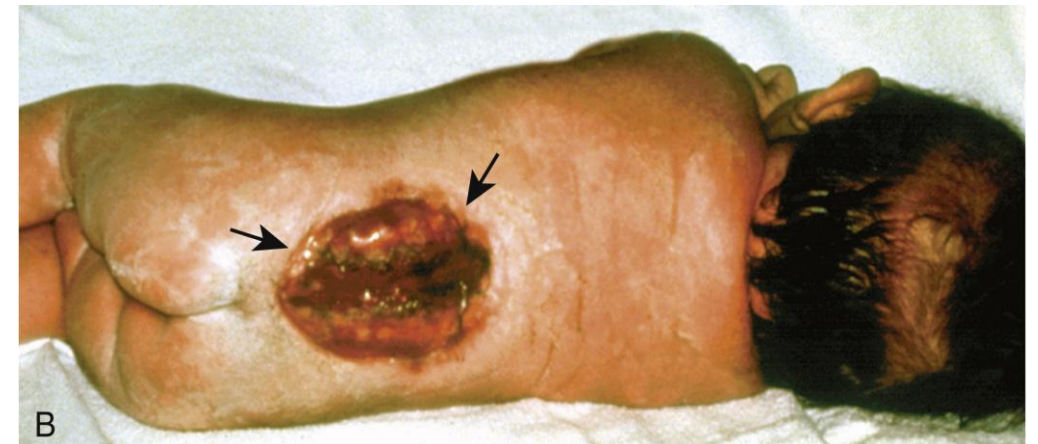
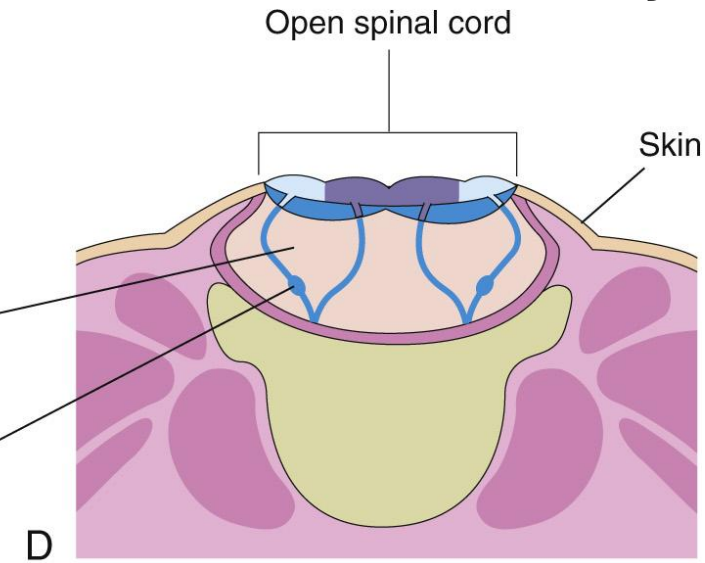
Cystica with meningocele

Bulge is seen over the defect
Contains meninges, cord & nerves



Cystica with myeloschisis

Skin and bony defect with "open" spinal cord, seen as a mass of neural tissue



Watch these videos for visual representation

- <https://premiumbasicsciences.lwwhealthlibrary.com/multimedia/player.aspx?MultimediaID=19450545>